

## **Chapter 4**

### **Project Plan Development**

The aim of plan development is to produce a usable, flexible, consistent, and logical document that will guide work or activities of the project. In addition, project plan offers a control mechanism for coordinating changes across the entire project. It is worth noting that project planning is an iteration process.

A first draft of the project plan is developed based on business case plus any other information as it becomes available. Historical from past projects could provide a useful resource for understanding how these project plans fared in terms of accuracy plus completeness of other estimates. They can also serve as source for drawing upon new ideas plus lessons learned. Also, policies and procedures of the organisation must be taken into account when developing the project plan e.g. formal accounting procedures may have to be followed for disbursement of funds for such things as travels, training or payments to vendors. On the other hand, an organisation might either have formal or informal policies for things such as hiring or firing employees or conducting performance and merit reviews.

Internal project teams may be familiar with these organisational policies while outside consultants may have to learn them as they go along. Irrespective of whether a project team is internal or external to the organisation, it is vital that project manager and team learn, understand and follow these policies since they can impact the project plan estimates.

Various constraints and assumptions must also be taken into consideration plus documented when developing project plan. Constraints are things that can limit the project and can normally have an impact on scope, schedule, budget or quality e.g. a project may have to be completed by a specific date or within a predefined budget

On the other hand, assumptions can be construed of as things that must go right in order for project plan to be completed as planned. Assumptions can be for instance skilled and experienced programmer being available by a specific date or a vendor delivering hardware and software in time for development activity to begin. Thus constraints and assumptions are closely related to risk.

Therefore the development of a risk management plan should be part of the project plan. The method for project planning is a crucial element for developing a project plan. All projects should follow a structured process. Various software tools, such as MS Project can be helpful in developing a project plan. Software tools nonetheless cannot create a project plan by itself. Thus, the project manager should involve various stakeholders throughout the planning process. Stakeholders could be managers or subject matter experts (SME) who can contribute valuable knowledge or expertise to refine the project plan. So project plan should also consider who will be needed, when they will be needed and how they will be needed to help create product of the project.

### **Project Plan Execution**

The aim of project planning process is to produce a document that can be executed in order to achieve project's measurable organisational value (MOV). It is vital to have a realistic and usable project plan because it will spend majority of its assigned resources in executing it.

Therefore, it is necessary that the plan be used not only to coordinate resources that will carry out certain scheduled activities but equally to measure the project's progress towards its goal.

Today, several organisations utilise some kind of project management software tool. Project management software tools not only help to produce and track a project's progress, but also acts as an IS for reporting project performance and making decisions. The project's product will directly determine the skills and knowledge areas required by project team members. So project manager must make sure that specific team members either have specific skills or knowledge coming into project or that they will acquire them in due course through training.

Project plan execution must have some sort of work authorisation system in place. Work authorisation is simply a way of approving or authorising team members to carry out a particular activity or group of related activities to certify that the right things are executed in the appropriate sequence. Work authorisation system can be formal or informal based on size and complexity of project. In smaller projects, work authorisation system could be nothing more than a project manager giving a project team verbal approval to start working on a specific activity outlined in the project plan.

While on the other hand, activities on larger, more complex projects may require a more formal approval since each team member may be working on a piece of application system. Thus, their activities may depend upon those of someone else or other group. Project manager must have a bigger picture in mind plus particular activities must be verified as being complete before others can start e.g. one set of activities for IT application system may be the gathering and documenting of requirements during system analysis phase. Many individuals or groups may work on this activity together. Design and programming activities should not begin until requirements are completed plus verified. Otherwise time and resources will be wasted if changes have to be made later.

Experience has proven that the cost of making changes or rectifying errors in later stages of a project is more expensive. Status review meetings are a useful tool for coordinating project processes and activities. These meetings are frequently scheduled meetings that project manager and team members have with key stakeholders. The aim of these meetings is to keep everyone informed as to the status of project. Project status meetings can be formal or informal and can include different levels of stakeholders. Regularly scheduled status meetings not only keep everyone informed, but equally help focus project team's attention on meeting key deadlines for deliverables. Meeting with project stakeholders tend to go on more cordially when project is progressing as planned.

### **Work Breakdown Structure**

As part of a project scope definition process, many tools and techniques were introduced e.g. deliverable definition table (DDT) and deliverable structure chart (DSC) which identify the deliverables that must be provided by project team. Once project's scope is identified, the next step is to define activities or tasks the team must do to fulfil scope deliverables requirements.

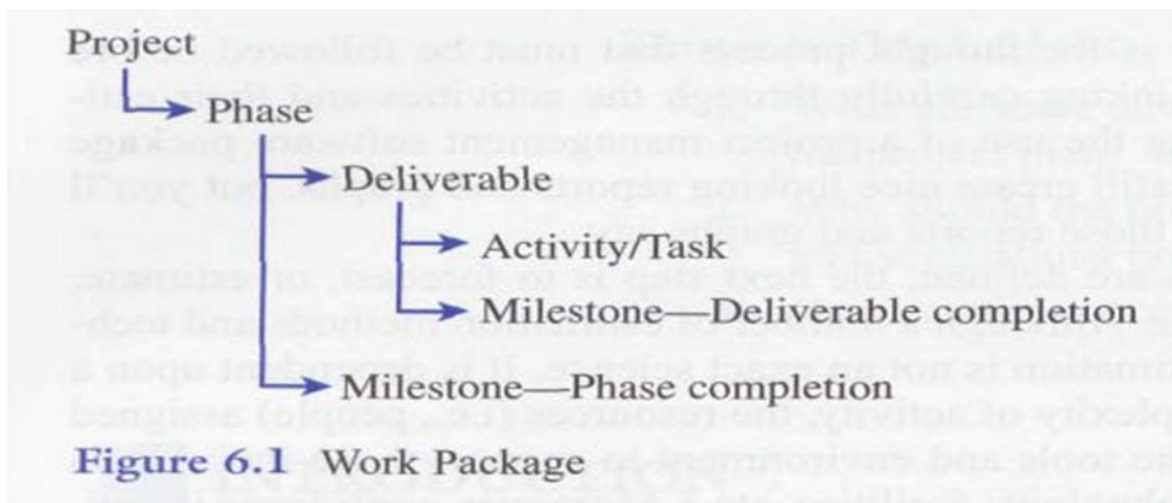
Work breakdown structure (WBS) is a useful tool for developing the project plan and links project's scope to the schedule and budget. According to Haugan (2002), the WBS represents a logical decomposition of the work to be performed and focuses on how the product, service or result is

naturally subdivided. It is an outline of what work is to be performed. WBS offers a framework for developing a strategic plan to structure project work.

PMBOK initially defined WBS as a “deliverable-oriented hierarchy”. But much debate and confusion has existed as to what a WBS should look like and how one should be built. As such, the Project Management Institute formed a committee to recommend standards for WBS. That committee recommended that no arbitrary limits should be mandated since WBS needs to be flexible. Thus, WBS can be deployed in different ways based on the needs of project manager and team.

### Work Packages

Work break structure (WBS) subdivides the project into smaller components and more manageable units of work called packages. Work packages provide a logical basis for defining project activities and assigning resources to those activities so that all the project work is identified (Haugan, 2002). Work package makes it possible to develop a project plan, schedule and budget plus then later monitor the project progress as depicted in Figure 6.1.



Marchewka, 2003

As demonstrated in Fig 6.1, work package may be viewed as a hierarchy which starts with the project itself. The project is then divided into phases with each phase having one or more deliverables as defined in the deliverable definition table and deliverable structure chart. More specifically, each phase should provide at least one specific deliverable i.e. a tangible and verifiable piece of work. Consequently, activities or tasks are identified in order to produce project's deliverables.

### Deliverables and Milestones

One departure from most conventional views of a WBS is inclusion of milestones.

A milestone is a significant event or an achievement that provides evidence indicating that a deliverable has been completed or a phase is formally over. Deliverables and milestones are closely linked but they are not the same thing. Deliverables may include such things as presentations, or reports, plans, prototypes and final application system while on the other hand, a milestone must

focus on an achievement e.g. a deliverable may be a prototype of the user interface but a milestone would be a stakeholder's formal approval of the user interface.

Only the formal acceptance of the user interface by the project sponsor would permit the project team to move onto the next phase of the project. In theory, if a project team succeeds in meeting all its scheduled milestones, then the project should be completed as planned.

Milestones equally offer many other advantages:

- First, milestones can keep the project team focused. It's much easier to concentrate your attention and efforts on a series of smaller short-term deliverables than on a single much larger deliverable scheduled for completion into the future.
- Equally, if milestones are realistic, they can motivate project team if their achievement is regarded as success. If attaining a milestone signifies an important event, then the team should take delight in the success before gearing up for the next milestone.
- Milestones equally minimise the risk of a project. Passing of a milestone, especially a phase milestone should offer an opportunity to review the progress of the project. Additional resources be committed at the successful completion of each milestone while appropriate plans and steps should be taken if project can't meet its milestones. Most often, a significant risk associated with IT projects is reliance on new technology or unique applications of technology.
- Milestones can also provide a mechanism for quality control e.g. simply providing the user with an interface doesn't guarantee that it will be acceptable to them. Thus, completion of user interface deliverable should culminate only with their assistance otherwise, the team will be forced to make revisions. In short, deliverable must not only be done, but must be done right.

### Developing the Work Breakdown Structure (WBS)

Developing the WBS may need several versions until everyone is comfortable and confident that all of the work activities have been included. It's also a good idea to involve those who will be doing the work since they probably know what has to be done better than anyone else. WBS can be quite loaded, depending upon the nature and size of project e.g. let us consider an electronic commerce project and if we focus on only one portion of the project – creating a document called test result report. Figure 6.2 depicts the Deliverable Structure Chart (DSC).

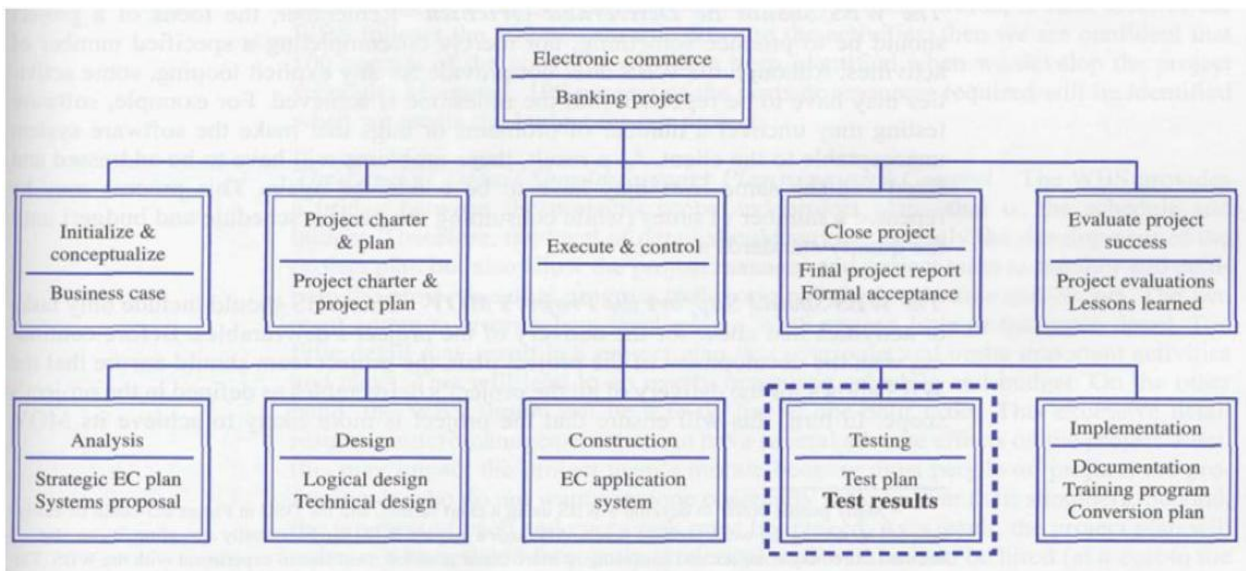


Figure 6.2 Deliverable Structure Chart (DSC) for EC Example

Marchewka, 2003

As we can see, 2 deliverables – the test plan and test report are to be completed and delivered during the testing phase of the project. DSC defines the phases and deliverables for our project.

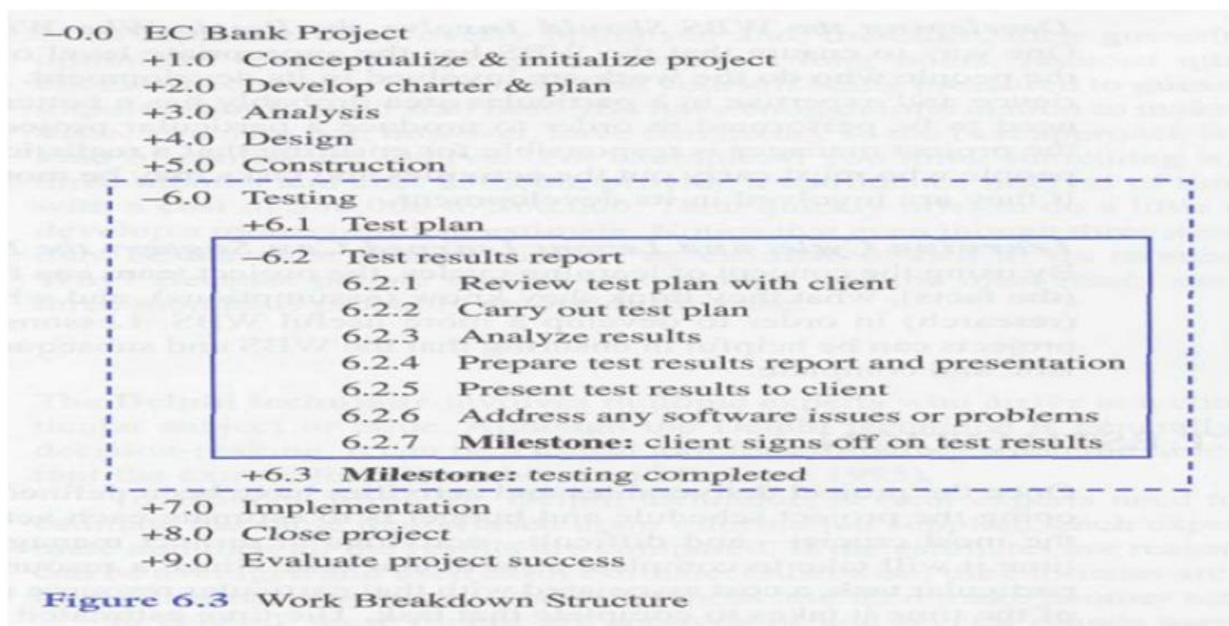
The next step is to develop sets of work packages for each of the phases and deliverable. After a team meeting, several activities that need to be done in order to produce the test results documents were identified and discussed:

- Review test plan with client in order that key stakeholders are clear as to what is being tested
- How these tests will be conducted and when they will be carried out

Review may be done as a courtesy or because specific help from client's organisation is needed and as such, must inform when that help/support will be required. After the client has been informed that system will be tested, tests outlined in the test plan are simply carried out.

Once test results have been collected, there is the need to analyse them. After analysing them, there is the need to summarise them in report form and presented to the client. If all goes well, then client will approve or sign off on test results. We can move on to the implementation phase of the project.

If all has not gone well, we will need to address and fix any problems. Bear in mind that the test phase is not complete simply because we have developed a test plan plus created a test report. The client will sign off on test results only if system meets certain predetermined quality standards. Fig 6.3 depicts a WBS detailing only the testing phase of a project.



As you can see, WBS implements the concept of a work package for project phase, deliverables, tasks/activity, and milestone components that were depicted in Fig 6.1. This particular WBS follows an outline format with a commonly used decimal numbering system that permits for continuing levels of detail. If a software package is deployed to create the WBS, signs in front of each item can either hide or show the details.

Skills to develop a useful WBS normally evolve over time with practice and experience. Thus, everyone experienced or not should bear in mind the following points when developing a WBS: